

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Geometric Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The general term

1. 1701 is the n th term of the G.P. 7, 21, 63, ... Find (a) n (b) the sum of these n terms.

ICSE 2025, 3 marks

SOLUTION

(a) $a = 7, r = 3. 7 \times 3^{n-1} = 1701$, so $3^{n-1} = 243 = 3^5$ and $n = 6$.

(b) $S_6 = \frac{a(r^6 - 1)}{r - 1} = \frac{7(729 - 1)}{2} = 7 \times 364$

Answer: (a) $n = 6$ (b) sum = 2548

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